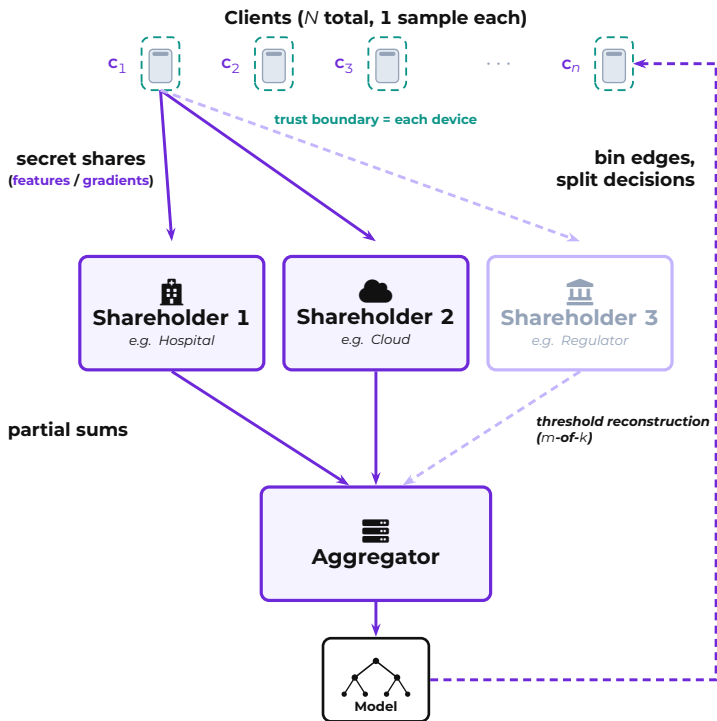




$\mathbf{x}$ = feature value $\mathbf{g}$ = gradient $\mathbf{h}$ = hessian

Phase 1: Statistics

Clients share x_i, x_i^2

Aggregator computes:

$$\mu = \frac{\sum x}{N}, \quad \sigma^2 = \frac{\sum x^2}{N} - \mu^2$$

Bins: $B+2$ edges at equal-probability quantiles of $\mathcal{N}(\mu, \sigma^2)$

Phase 2: Gradients

Clients share g_i, h_i per bin

Per bin b : $G_b = \sum g_i, \quad H_b = \sum h_i$

Best split: $\max_b \frac{G_b^2}{H_b + \lambda}$

Privacy Guarantees

- Shareholders see random shares
- Aggregator sees only sums
- No individual values revealed

All computations use only **sums** of secret-shared values